

LINEARITY OF NON-ABELIAN TENSOR SQUARES: A COUNTEREXAMPLE AND THE BRAID GROUPS

GROUPTHEORY50

ABSTRACT. We answer both parts of Kurovka Problem 19.9. First, for $r \geq 4$ the finitely generated linear group $\mathrm{Sp}_{2r}(\mathbb{Z})$ has a non-linear non-abelian tensor square, by combining the Brown–Loday description for perfect groups with Deligne’s non-residually finite universal central extension. Second, for every $n > 3$ the tensor square $B_n \otimes B_n$ of the braid group is linear. The latter follows from the homology map $H_2(B_n; \mathbb{Z}) \rightarrow H_2(S_n; \mathbb{Z})$ and the standard splitting of the tensor square when the abelianization has no 2-torsion.

1. INTRODUCTION

Brown and Loday introduced the non-abelian tensor product and related it to low-dimensional homotopy and group homology [5]. Bardakov, Lavrenov and Neshchadim showed that linearity of G alone does not force linearity of $G \otimes G$, using non-finitely-generated examples, and isolated the finitely generated case as an open problem [3]. This became Kurovka Problem 19.9: if G is finitely generated and linear, must $G \otimes G$ be linear, and in particular what happens for braid groups B_n , $n > 3$ [7]?

We show that the general answer is negative but the braid-group answer is positive.

2. A FINITELY GENERATED COUNTEREXAMPLE

For a group G there is an exact sequence

$$H_2(G; \mathbb{Z}) \longrightarrow G \wedge G \xrightarrow{\kappa} G' \longrightarrow 1.$$

If G is perfect, then $G \otimes G = G \wedge G$ and this is the universal central extension of G [5].

Theorem 2.1. *There is a finitely generated linear group G such that $G \otimes G$ is not linear.*

Proof. Take

$$G = \mathrm{Sp}_{2r}(\mathbb{Z}), \quad r \geq 4.$$

This group is finitely generated, perfect, and linear. In this range the pullback of the universal cover of $\mathrm{Sp}_{2r}(\mathbb{R})$ is the universal central extension

$$1 \longrightarrow \mathbb{Z} \longrightarrow \tilde{G} \longrightarrow G \longrightarrow 1$$

[1]. Deligne proved that every finite quotient of $\tilde{G}$ kills a nonzero multiple of the central generator; in particular $\tilde{G}$ is not residually finite [6].

Since G is perfect, Brown–Loday gives $G \otimes G \cong \tilde{G}$. A finitely generated linear group is residually finite by Mal’cev’s theorem. Therefore $G \otimes G$ cannot be linear. $\square$

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3. THE BRAID GROUPS

Let $n \geq 4$ and let

$$\beta : B_n \twoheadrightarrow S_n$$

be the standard epimorphism with kernel the pure braid group P_n . We use

$$B_n^{\text{ab}} \cong \mathbb{Z}, \quad H_2(B_n; \mathbb{Z}) \cong C_2, \quad H_2(S_n; \mathbb{Z}) \cong C_2.$$

Lemma 3.1. *The induced map*

$$\beta_* : H_2(B_n; \mathbb{Z}) \longrightarrow H_2(S_n; \mathbb{Z})$$

is an isomorphism.

Proof. The five-term sequence for

$$1 \rightarrow P_n \rightarrow B_n \rightarrow S_n \rightarrow 1$$

contains

$$H_2(B_n) \rightarrow H_2(S_n) \rightarrow H_1(P_n)_{S_n} \rightarrow H_1(B_n) \rightarrow H_1(S_n) \rightarrow 0.$$

The coinvariants $H_1(P_n)_{S_n}$ are infinite cyclic because S_n is transitive on the standard pure-braid generators A_{ij} . The class of $A_{12} = \sigma_1^2$ maps to twice the generator of $H_1(B_n) \cong \mathbb{Z}$. Hence the map $H_1(P_n)_{S_n} \rightarrow H_1(B_n)$ is injective. Therefore $H_2(B_n) \rightarrow H_2(S_n)$ is surjective, and since both groups have order two, it is an isomorphism. $\square$

Put $A_n = S'_n = \text{Alt}_n$ and define the pullback

$$E_n = B'_n \times_{A_n} (S_n \wedge S_n).$$

Proposition 3.2. *The natural map $B_n \wedge B_n \rightarrow E_n$ is an isomorphism.*

Proof. Both groups are central extensions of B'_n . The induced map is the identity on the quotient, while on the kernels it is precisely the isomorphism of Lemma 3.1. The short five lemma gives the result. $\square$

Theorem 3.3. *For every $n > 3$, the non-abelian tensor square $B_n \otimes B_n$ is linear.*

Proof. The group $S_n \wedge S_n$ is finite. The group B'_n is linear because B_n is linear by the theorems of Bigelow and Krammer [4, 8]. Hence the pullback E_n , as a subgroup of $B'_n \times (S_n \wedge S_n)$, is linear. Proposition 3.2 therefore shows that $B_n \wedge B_n$ is linear.

Since $B_n^{\text{ab}} \cong \mathbb{Z}$ has no 2-torsion, the splitting theorem for non-abelian tensor squares gives

$$B_n \otimes B_n \cong \nabla(B_n) \times (B_n \wedge B_n), \quad \nabla(B_n) \cong \Gamma(\mathbb{Z}) \cong \mathbb{Z}$$

[2]. Thus $B_n \otimes B_n$ is a direct product of linear groups and is linear. $\square$

4. CONCLUSION

The two parts of Problem 19.9 exhibit genuinely different behavior: finite generation does not rescue tensor-square linearity in general, while braid groups retain linearity after passage to the non-abelian tensor square.

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